

Supplementary Material for “Certified Identifiability of Fractional ECG Harmonic Surrogates from Multi-Rate Records”

Complete Proofs and Certificate Derivations

Anonymous authors

Abstract

We establish the theory gates that precede real-data validation. The first result proves that two exact complex rate responses are necessary and sufficient to recover the order of a scalar fractional ECG harmonic surrogate with an unknown common complex coefficient and positive real damping. We then give an exact multi-rate reference-pair consistency test, prove that unrestricted rate-specific complex morphology destroys identifiability and that, under the stated two-rate known-damping interior conditions, any positive independent complex-disk morphology drift prevents singleton order recovery, and give a quadratic upper bound on Gram-margin collapse near real-separable response arrays. We then give a joint forward-disk feasible-set theorem, a gauge-reduced analytic nuisance-selector tube with an exact spectral margin and second-order path enclosure, replayable finite-record component bounds, a frozen-witness R2-to-T1 transfer theorem, and an exact output-set separation threshold for worst-case rate-window design. The complete exact-sample chain is implemented for a frozen synthetic functional, but the required acquisition and model-error primitives are not calibrated on the real records. The real-data result therefore remains not certifiable. These statements concern a declared surface-ECG surrogate, not internal cardiac states.

1 Model, units, and protocol

Fix one reference time $\tau_\star > 0$ before identification, put $\xi = t/\tau_\star$, and use the dimensionless model

$${}^C D_\xi^\alpha z(\xi) = -\lambda z(\xi) + \sum_{m \in \mathbb{Z}} q_m e^{im\theta(\xi)}. \quad (1)$$

Thus λ is dimensionless and q_m has the same measurement unit as z , independently of the candidate order. A finite-lower-terminal Caputo system is not assumed to possess an exact periodic orbit [6]. The frequency response below is the infinite-history (Weyl) harmonic branch, equivalently the asymptotic periodic component of the finite-terminal Caputo solution [7]; all initial-condition, prehistory, and finite-dwell terms are retained later as R2 residuals. With dimensionless frequency $\nu_{r,m} = m\tau_\star\Omega_r$, where Ω_r is the base angular heart rate in rad/s, one selected positive harmonic has response

$$Z_r = \frac{q}{\lambda + (i\nu_r)^\alpha}, \quad (i\nu)^\alpha = \nu^\alpha e^{i\pi\alpha/2}, \quad (2)$$

where

$$\alpha \in I = [\alpha_-, 1], \quad 0 < \alpha_- < 1, \quad \lambda \in \Lambda = [\lambda_-, \lambda_+] \subset \mathbb{R}_{>0}, \quad q \in \mathbb{C} \setminus \{0\}.$$

The coefficient q is effective in measurement units. A common nonzero complex lead gain is absorbed into q . For a real ECG, the corresponding negative harmonic is its conjugate and supplies no independent equation. Throughout, $\mathcal{K}_{\text{est}} \subset \mathbb{Z}$ denotes the full estimator basis, which may contain DC and conjugate negative harmonics, whereas $\mathcal{M} \subset \mathbb{N}_{>0}$ contains only the positive target harmonics passed to R1. Thus $\mathcal{M} \subseteq \mathcal{K}_{\text{est}}$; DC is an estimator nuisance and a negative coefficient is a conjugacy check, never an input to a selector that uses $\log \nu_{r,m}$.

Assumption S1 (Information boundary). Rates are known, positive, and distinct. Complex responses share one R-peak phase anchor, one causal preprocessing chain, the principal fractional branch, and common q, λ, α . There is no uncalibrated rate-specific gain, phase, or delay. Restitution bounds, when used, are calibrated independently of the identification record.

2 R0: minimal structural identifiability

Proposition S2 (One rate is insufficient). *For one response Z_1 , the order is not structurally identifiable.*

Proof. For every $\beta \in I$ and $\ell \in \Lambda$, the nonzero choice

$$q_{\beta, \ell} = Z_1[\ell + (i\nu_1)^\beta]$$

reproduces the same observation. □

For any ordered pair of nonzero responses, put

$$W_j = Z_j^{-1}, \quad D = W_2 - W_1,$$

and, whenever $D \neq 0$, define $C = W_1/D$.

Lemma S3 (Complete common-gauge invariant). *For an ordered nonzero response pair with $D \neq 0$, the quotient C is invariant under $(Z_1, Z_2) \mapsto (gZ_1, gZ_2)$, $g \in \mathbb{C} \setminus \{0\}$. It is complete: two such ordered pairs have the same C if and only if one is obtained from the other by one common nonzero complex gain.*

Proof. The inverse responses obey $W_1 = CD$ and $W_2 = (C + 1)D$. If a second pair has difference D' , then $W'_j = (D'/D)W_j$ and hence $Z'_j = (D/D')Z_j$. The converse follows directly from cancellation in $C = W_1/(W_2 - W_1)$. □

Theorem S4 (Quotient-space geometry). *For fixed $\varrho > 1$, on the enlarged structural domain before the compact priors $I \times \Lambda$ are imposed, the map*

$$\Phi : (\alpha, \kappa) \mapsto \frac{1 + \kappa e^{-i\pi\alpha/2}}{\varrho^\alpha - 1}$$

is one-to-one on $(0, 1] \times \mathbb{R}_{>0}$, and its range is

$$\mathcal{Q}_\varrho = \left\{ u + iv : u \geq \frac{1}{\varrho - 1}, v < 0 \right\}.$$

It is a smooth diffeomorphism from $(0, 1) \times \mathbb{R}_{>0}$ onto $\{u + iv : u > 1/(\varrho - 1), v < 0\}$, and extends one-to-one to $\alpha = 1$. In the interior its real Jacobian is

$$\det \frac{\partial(\operatorname{Re} C, \operatorname{Im} C)}{\partial(\alpha, \kappa)} = \frac{\varrho^\alpha \log \varrho \sin(\pi\alpha/2)}{(\varrho^\alpha - 1)^3} + \frac{\kappa\pi}{2(\varrho^\alpha - 1)^2} > 0. \quad (3)$$

Proof. Every image point has negative imaginary part; for $\alpha < 1$, its real part is strictly larger than $1/(\varrho - 1)$, whereas $\alpha = 1$ gives the boundary line. Conversely, for $v < 0$, define

$$H_{u+iv}(\beta) = (\varrho^\beta - 1)[u + v \cot(\pi\beta/2)].$$

The continuous equation $H_{u+iv}(\beta) = 1$ has a root in $(0, 1]$ whenever $(\varrho - 1)u \geq 1$: its limit at zero is negative and its value at one is at least one. The recovered

$$\kappa = -\frac{(\varrho^\alpha - 1)v}{\sin(\pi\alpha/2)}$$

is positive. If α is one root and $A_\gamma = \pi\gamma/2$, then direct substitution gives, for any candidate β ,

$$(\varrho^\alpha - 1)[H_{u+iv}(\beta) - 1] = \varrho^\beta - \varrho^\alpha + \kappa(\varrho^\beta - 1)\frac{\sin(A_\beta - A_\alpha)}{\sin A_\beta}.$$

Both terms have the sign of $\beta - \alpha$, so the root and its positive κ are unique. Lemma S3 shows that this is the complete common-gauge quotient. Direct differentiation gives (3); the inverse function theorem and the established bijection give the interior diffeomorphism. $\square$

Theorem S5 (Global two-rate recovery). *Let $0 < \nu_1 < \nu_2$, and suppose two exact complex responses satisfy (2). Define*

$$W_r = Z_r^{-1}, \quad D = W_2 - W_1, \quad C = \frac{W_1}{D}, \quad \varrho = \frac{\nu_2}{\nu_1} > 1.$$

Then $D \neq 0$, $\text{Im } C < 0$, and the true order is the unique $\beta \in (0, 1]$ satisfying

$$H_C(\beta) = 1, \quad H_C(\beta) = (\varrho^\beta - 1) \left[\text{Re } C + \text{Im } C \cot \left(\frac{\pi\beta}{2} \right) \right]. \quad (4)$$

After recovering α ,

$$q = \frac{e^{i\pi\alpha/2}(\nu_2^\alpha - \nu_1^\alpha)}{D}, \quad (5)$$

$$\lambda = \nu_1^\alpha e^{i\pi\alpha/2} [C(\varrho^\alpha - 1) - 1] \in \mathbb{R}_{>0}. \quad (6)$$

Proof. Inverse-response subtraction gives

$$D = \frac{e^{i\pi\alpha/2}(\nu_2^\alpha - \nu_1^\alpha)}{q} \neq 0.$$

With $\kappa = \lambda/\nu_1^\alpha > 0$ and $A = \pi\alpha/2$,

$$C = \frac{1 + \kappa e^{-iA}}{\varrho^\alpha - 1}, \quad (7)$$

so $\text{Im } C < 0$. A candidate β first fixes q_β through the difference equation. Substitution into the first response gives

$$\lambda_\beta = \nu_1^\beta e^{i\pi\beta/2} [C(\varrho^\beta - 1) - 1].$$

Writing $C = u + iv$, $t = \varrho^\beta - 1$, and $B = \pi\beta/2$,

$$\text{Im } \lambda_\beta = 0 \iff (tu - 1) \sin B + tv \cos B = 0 \iff H_C(\beta) = 1.$$

Substitute (7) and use

$$\cos A - \sin A \cot B = \frac{\sin(B - A)}{\sin B}.$$

The candidate equation becomes

$$\varrho^\beta - \varrho^\alpha + \kappa(\varrho^\beta - 1)\frac{\sin(B - A)}{\sin B} = 0. \quad (8)$$

Both terms are positive when $\beta > \alpha$ and negative when $\beta < \alpha$; hence $\beta = \alpha$ is the unique solution. At a root,

$$\text{Re } \lambda_\beta = -\nu_1^\beta \frac{t \text{Im } C}{\sin B} > 0,$$

and $D \neq 0$ ensures $q_\beta \neq 0$. $\square$

Corollary S6 (Strict minimum and quotient invariance). *Under Assumption S1, two distinct rates are the strict structural minimum. Under $Z_r \mapsto gZ_r$, $g \in \mathbb{C} \setminus \{0\}$, C, α, λ are invariant, while (5) recovers gq_{physical} . Two exact responses can already exclude the declared model when their quotient lies outside the physical image or the compact priors. A third rate is the minimum that adds a redundant cross-rate consistency equation after an admissible reference pair has been inverted.*

Theorem S7 (Exact reference-pair consistency). *Consider the zero-drift single-harmonic model at $0 < \nu_1 < \dots < \nu_R$, $R \geq 2$, with exact nonzero responses. Put*

$$W_r = Z_r^{-1}, \quad C_{1r} = \frac{W_1}{W_r - W_1}, \quad \varrho_r = \frac{\nu_r}{\nu_1}.$$

Assume each $W_r \neq W_1$ and $C_{1r} \in \mathcal{Q}_{\varrho_r}$. Let $(\alpha_r, \kappa_r) = \Phi_{\varrho_r}^{-1}(C_{1r})$ be the unique inverse from Theorem S4. On the enlarged structural domain $(0, 1] \times \mathbb{R}_{>0} \times (\mathbb{C} \setminus \{0\})$, all responses arise from one common (α, λ, q) if and only if

$$(\alpha_r, \kappa_r) = (\alpha_*, \kappa_*) \quad \text{for every } r = 2, \dots, R.$$

Consistency with the declared compact model additionally holds if and only if $\alpha_ \in I$ and $\kappa_* \nu_1^{\alpha_*} \in \Lambda$. In either structural case*

$$\lambda = \kappa_* \nu_1^{\alpha_*}, \quad q = \frac{\lambda + (i\nu_1)^{\alpha_*}}{W_1}.$$

Proof. Necessity follows by substituting the common model into every reference pair. For sufficiency, the inverse quotient identity gives

$$\frac{W_1}{C_{1r}} = \frac{(i\nu_r)^{\alpha_*} - (i\nu_1)^{\alpha_*}}{q}.$$

Since the left side is $W_r - W_1$, adding $W_1 = [\lambda + (i\nu_1)^{\alpha_*}]/q$ proves the model equation for every rate. $\square$

Corollary S8 (Geometric three-rate cross-ratio). *In Theorem S7, take $\nu_2 = \rho\nu_1$ and $\nu_3 = \rho^2\nu_1$, $\rho > 1$, and define*

$$Q = \frac{W_3 - W_1}{W_2 - W_1}.$$

Every response triple in the declared zero-drift model satisfies the damping- and morphology-free identity

$$Q = 1 + \rho^\alpha \in \mathbb{R}, \quad \alpha = \frac{\log(Q - 1)}{\log \rho}.$$

Conversely, an admissible reference pair $(1, 2)$ with inverse (α_, κ_*) extends to the third response if and only if*

$$Q = 1 + \rho^{\alpha_*}.$$

Together with $\alpha_ \in I$ and $\kappa_* \nu_1^{\alpha_*} \in \Lambda$, this is an exact three-rate model-consistency test. Thus the third rate is a genuine redundant equation, not another fitted nuisance parameter.*

Proof. Under the model,

$$W_r - W_1 = \frac{(i\nu_r)^\alpha - (i\nu_1)^\alpha}{q}.$$

The common factors $q^{-1}(i\nu_1)^\alpha$ cancel, and

$$Q = \frac{\rho^{2\alpha} - 1}{\rho^\alpha - 1} = 1 + \rho^\alpha.$$

The logarithmic inverse is unique because $\rho > 1$. Conversely, the reference pair fixes α_*, λ, q by Theorem S7; the displayed cross-ratio identity is exactly the remaining equation $W_3 - W_1$. $\square$

Corollary S9 (Exact multi-harmonic consistency). *For positive target harmonics in the zero-drift model, apply Theorem S7 within each harmonic. A common α_* , λ_* and harmonic-specific q_m exist if and only if the reference-pair inverses satisfy*

$$\alpha_{r,m} = \alpha_*, \quad \kappa_{r,m} = \kappa_m \quad \text{for every } r, m,$$

and

$$\kappa_m \nu_{1,m}^{\alpha_*} = \lambda_* \quad \text{is independent of } m.$$

This test does not apply unchanged to the later rate-specific damping-drift model. With response disks, pairwise centre inversions are not a certificate; the complete joint set must be used.

Proposition S10 (Rate-specific complex nuisance destroys identifiability). *Let all $Z_r \neq 0$, and remove every annulus or cross-rate constraint on nonzero $q_r \in \mathbb{C}$. In*

$$Z_r = \frac{q_r}{\lambda + (i\nu_r)^\alpha},$$

every candidate $(\beta, \ell) \in I \times \mathbb{R}_{>0}$ fits every finite number of rates by choosing

$$q_r = Z_r[\ell + (i\nu_r)^\beta].$$

The same conclusion holds when an arbitrary unknown nonzero complex gain is allowed at every rate. Thus per-window complex normalization removes all order information carried by the common gauge.

Proof. The displayed construction is nonzero and reproduces each response exactly. An unrestricted rate-specific gain is algebraically equivalent to such an unrestricted numerator. $\square$

Corollary S11 (Quotient imaginary-axis separation under compact priors). *For a fixed pair, $I = [\alpha_-, 1]$, and $\lambda \geq \lambda_- > 0$,*

$$-\operatorname{Im} C \geq \gamma_C := \lambda_- \min_{\alpha \in I} \frac{\nu_1^{-\alpha} \sin(\pi\alpha/2)}{\varrho^\alpha - 1} > 0.$$

This is only separation from the quotient real axis. It does not control $W_2 - W_1$, and it does not remove the nonstrict real-part boundary at $\alpha = 1$.

For stable computation after α is isolated, the equivalent real recovery

$$\kappa = -\frac{(\varrho^\alpha - 1) \operatorname{Im} C}{\sin(\pi\alpha/2)} > 0, \quad \lambda = \nu_1^\alpha \kappa$$

avoids accepting a nearly real complex value from (6); the recovered λ must also be checked against Λ .

Remark S12 (Boundary and computation).

$$\lim_{\beta \rightarrow 0^+} H_C(\beta) = \frac{2 \operatorname{Im} C \log \varrho}{\pi} < 0.$$

At $\alpha = 0$, $D = 0$; no uniform conditioning exists as $\alpha_- \rightarrow 0$ or $\varrho \rightarrow 1$. Although model-consistent data have one root, H_C need not be monotone on all of I . Nonideal data therefore require complete root isolation or set inversion, not an unverified monotone inverse. The endpoint $\alpha = 1$ is admissible and is not rejected merely because it lies on the boundary.

3 R1: joint forward-disk set

Assume deterministic response disks

$$|Z_{r,m} - \tilde{Z}_{r,m}| \leq \epsilon_{r,m}. \quad (9)$$

For every retained harmonic, pre-register the compact annulus

$$\mathcal{E}_m = \{q \in \mathbb{C} : q_{\min,m} \leq |q| \leq q_{\max,m}\}, \quad 0 < q_{\min,m} < q_{\max,m} < \infty.$$

For $(\beta, \ell) \in I \times \Lambda$, let

$$d_{r,m}(\beta, \ell) = \ell + (i\nu_{r,m})^\beta$$

and

$$\mathcal{Q}_{r,m}(\beta, \ell) = \mathcal{D}(d_{r,m}\tilde{Z}_{r,m}, |d_{r,m}|\epsilon_{r,m}). \quad (10)$$

Theorem S13 (Fixed-candidate forward-disk equivalence). *For fixed (β, ℓ) , there is one $q_m \in \mathcal{E}_m$, shared by all rates, satisfying*

$$\left| \tilde{Z}_{r,m} - \frac{q_m}{d_{r,m}(\beta, \ell)} \right| \leq \epsilon_{r,m} \quad \forall r$$

if and only if

$$\left(\bigcap_r \mathcal{Q}_{r,m}(\beta, \ell) \right) \cap \mathcal{E}_m \neq \emptyset. \quad (11)$$

The joint zero-drift order set is

$$\mathcal{A}_{\text{joint}} = \text{proj}_\beta \left\{ (\beta, \ell) \in I \times \Lambda : \left(\bigcap_r \mathcal{Q}_{r,m}(\beta, \ell) \right) \cap \mathcal{E}_m \neq \emptyset \quad \forall m \in \mathcal{M} \right\}. \quad (12)$$

Proof. Because $d_{r,m} \neq 0$,

$$\left| \tilde{Z}_{r,m} - q_m/d_{r,m} \right| \leq \epsilon_{r,m} \iff |d_{r,m}\tilde{Z}_{r,m} - q_m| \leq |d_{r,m}|\epsilon_{r,m}.$$

The right side is membership in $\mathcal{Q}_{r,m}$. A shared coefficient gives the intersection, and the annulus imposes compact nondegeneracy. $\square$

Corollary S14 (Independent complex morphology restitution). *If $q_{r,m} = \bar{q}_m + \delta q_{r,m}$, $\bar{q}_m \in \mathcal{E}_m$, and each $|\delta q_{r,m}| \leq \rho_{r,m}^q$ is an independent complex-disk variable, then Theorem S13 remains exact after replacing the radius in (10) by*

$$|d_{r,m}|\epsilon_{r,m} + \rho_{r,m}^q.$$

Correlated, real-only, or cross-harmonic restitution must retain its shared variables; disk inflation is then only an outer enclosure.

Theorem S15 (Positive independent morphology drift precludes point order). *Fix two rates and a known $\lambda_0 > 0$. Suppose exact zero-drift responses are generated at $(\alpha_0, \lambda_0, q_0)$, where the relevant single-harmonic morphology domain is the compact annulus $\mathcal{E} = \{q : q_{\min} \leq |q| \leq q_{\max}\}$, $0 < q_{\min} < q_{\max} < \infty$, and $q_0 \in \text{int } \mathcal{E}$. Allow independent morphology disks of radii $\rho_1, \rho_2 \geq 0$ with $\rho_1 + \rho_2 > 0$. Fix $h_0 > 0$ and put*

$$c_r(\beta) = Z_r[\lambda_0 + (i\nu_r)^\beta], \quad F(\beta) = c_1(\beta) - c_2(\beta),$$

and, on a compact relative neighborhood $J = I \cap [\alpha_0 - h_0, \alpha_0 + h_0]$, define

$$q_w(\beta) = \frac{\rho_2 c_1(\beta) + \rho_1 c_2(\beta)}{\rho_1 + \rho_2}, \quad L_F = \sup_J |F'|, \quad L_w = \sup_J |q'_w|.$$

With division by zero interpreted as $+\infty$, every $\beta \in I$ satisfying

$$|\beta - \alpha_0| \leq h := \min \left\{ h_0, \frac{\rho_1 + \rho_2}{L_F}, \frac{\text{dist}(q_0, \mathcal{E}^c)}{L_w} \right\}$$

is feasible. In particular, the order projection contains a nondegenerate relative neighborhood of α_0 , one-sided when α_0 is an endpoint. Enlarging the damping prior while retaining λ_0 , or adding response disks centred at the same exact responses with nonnegative radii, can only enlarge this set.

Proof. At the truth, $c_1(\alpha_0) = c_2(\alpha_0) = q_0$. The mean-value bound gives $|F(\beta)| \leq L_F |\beta - \alpha_0| \leq \rho_1 + \rho_2$. The weighted centre satisfies

$$|q_w - c_1| = \frac{\rho_1}{\rho_1 + \rho_2} |F| \leq \rho_1, \quad |q_w - c_2| \leq \rho_2,$$

and $|q_w(\beta) - q_0| \leq L_w |\beta - \alpha_0|$ keeps it in $\mathcal{E}$. Thus q_w is a common witness. Without the annulus, two-disk intersection is exactly equivalent to $|c_1 - c_2| \leq \rho_1 + \rho_2$; with the annulus the exact condition remains the three-way intersection and the distance inequality alone is not sufficient. $\square$

Definition S16 (Shared damping restitution). Let

$$\lambda_r = \bar{\lambda} + \delta\lambda_r, \quad \bar{\lambda} \in \bar{\Lambda}, \quad |\delta\lambda_r| \leq \rho_r^\lambda, \quad \lambda_r \in \Lambda_r \subset \mathbb{R}_{>0}.$$

The same real $\delta\lambda_r$ is shared across all harmonics at rate r . If $\bar{\lambda}$ needs a unique nominal interpretation, impose a zero-mean or anchored drift gauge.

Corollary S17 (Exact joint set with declared restitution). *Let*

$$d_{r,m}^{\text{dr}} = \bar{\lambda} + \delta\lambda_r + (i\nu_{r,m})^\beta, \quad \mathcal{Q}_{r,m}^{\text{dr}} = \mathcal{D}(d_{r,m}^{\text{dr}} \tilde{Z}_{r,m}, |d_{r,m}^{\text{dr}}| \epsilon_{r,m} + \rho_{r,m}^q).$$

Then the exact order set under independent complex morphology disks and shared real rate-indexed damping drift is

$$\text{proj}_\beta \left\{ (\beta, \bar{\lambda}, \delta\lambda) : \begin{array}{l} \beta \in I, \bar{\lambda} \in \bar{\Lambda}, |\delta\lambda_r| \leq \rho_r^\lambda, \bar{\lambda} + \delta\lambda_r \in \Lambda_r, \\ (\bigcap_r \mathcal{Q}_{r,m}^{\text{dr}}) \cap \mathcal{E}_m \neq \emptyset \quad \forall m \in \mathcal{M} \end{array} \right\}.$$

A damping gauge is required to interpret $\bar{\lambda}$ uniquely, but not to form the conservative projection onto β .

Remark S18 (Certified algorithm boundary). Fixed-parameter disk intersection with an annulus has finite planar geometry. The current `certified_set.py` implementation embeds numerical inputs in containing Python-FLINT Arb/ACB balls, preserves immutable rate identifiers and the anchored damping gauge, and uses three fast outer-exclusion tests followed by a bounded Cartesian q -box cover. Ordinary floating-point geometry may only propose an inner witness; Arb/ACB inequalities verify it before inclusion is issued. Thus every issued outer exclusion or uniform inner inclusion is a machine certificate. Precision, q -box, branch, or leaf-budget exhaustion remains unknown, so this certified core is not a guaranteed finite exact projector. The companion `tube_certificate.py` implements Proposition S28 and Theorem S32: it uses an exact-rational Gram inverse, Arb path inequalities, sealed per-interval evidence, and a replayed alpha-only binary cover. It issues only robust inner, closed inner, or unknown results.

Assumption S19 (Certification coordinates). Every rate has an immutable identifier shared by all harmonics, and harmonic frequencies are derived from one rate-indexed base-frequency vector rather than independently sorted arrays. Rate-dependent damping is written $\lambda_r = \bar{\lambda} + \delta\lambda_r$, with one preregistered reference rate satisfying $\delta\lambda_{r_0} = 0$. This anchor removes the translation gauge

between $\bar{\lambda}$ and the offsets. Before any norm or Gram margin is evaluated, the protocol freezes a dimensionless numerical column scaling for the mixed damping and morphology coordinates; every norm below acts on that preregistered representation. All numerical inputs are embedded in containing real or complex balls.

Lemma S20 (Anchored damping gauge is lossless). *For a fixed reference rate r_0 , the map*

$$(\lambda_r)_r \mapsto (\bar{\lambda} = \lambda_{r_0}, \quad \delta\lambda_r = \lambda_r - \lambda_{r_0})_r$$

is a bijection between positive physical rate dampings and anchored coordinates satisfying $\delta\lambda_{r_0} = 0$ and $\bar{\lambda} + \delta\lambda_r > 0$. It leaves every denominator $d_{r,m}$, hence the feasible set and its order projection, unchanged.

Proof. The displayed construction gives the anchored coordinates of any physical damping vector. Conversely, $\lambda_r = \bar{\lambda} + \delta\lambda_r$ is the unique inverse, because the anchor forces $\bar{\lambda} = \lambda_{r_0}$. Substitution preserves every model denominator exactly. $\square$

Lemma S21 (Rank test for the fully free damping centre map). *Suppose $R, M \geq 1$ and every retained centre $\tilde{Z}_{r,m}$ is nonzero. In physical damping coordinates, with all R physical rate dampings free, let the real matrix B_{full} represent the linear map*

$$(a_1, \dots, a_R, b_1, \dots, b_M) \mapsto (b_m - \tilde{Z}_{r,m} a_r)_{r,m}, \quad a_r \in \mathbb{R}, \quad b_m \in \mathbb{C}.$$

The complex outputs are stacked by real and imaginary parts. For any reference rate r_0 , B_{full} has full column rank if and only if there do not exist nonzero real numbers κ_r such that

$$\tilde{Z}_{r,m} = \kappa_r \tilde{Z}_{r_0,m} \quad \text{for every } r, m. \quad (13)$$

The same condition holds after the invertible anchored coordinate change of Assumption S19. In particular, $2RM \geq R + 2M$ is necessary in this fully free case.

Proof. A null vector satisfies $b_m = \tilde{Z}_{r,m} a_r$ for every r, m . If one a_r vanishes, nonzero response centres force every b_m , and then every a_r , to vanish. Hence a nonzero null vector has all $a_r \neq 0$, and division by the equation at r_0 gives $\tilde{Z}_{r,m}/\tilde{Z}_{r_0,m} = a_{r_0}/a_r$, which is real and independent of m . Conversely, if (13) holds, $a_{r_0} = 1$, $a_r = 1/\kappa_r$, and $b_m = \tilde{Z}_{r_0,m}$ form a nonzero null vector. The anchored-to-physical damping map in Lemma S20 is invertible, so it cannot change rank. The row-count condition follows immediately. $\square$

Corollary S22 (Exact elimination of fixed damping coordinates). *Before constructing a selector, fix every zero-width damping coordinate at its declared value and absorb its contribution into the order-dependent centre vector. If p damping coordinates remain free, the resulting residual has the exact reduced form*

$$B_J y - g_J(\beta), \quad y \in \mathbb{R}^{p+2M}.$$

Strict convexity requires only that B_J have full column rank; hence the necessary row-count condition is $2RM \geq p + 2M$. This elimination changes neither the feasible fibres nor their order projection. The real-separability criterion of Lemma S21 is the $p = R$ special case; for a general free-coordinate set J , rank is checked on B_J itself.

Proof. Substitution of a fixed coordinate is an exact affine restriction. Moving its known term to $g_J(\beta)$ leaves every residual value unchanged on that restriction. The row-count condition follows from the reduced dimensions; full column rank of B_J remains a separate condition to be checked. $\square$

Theorem S23 (Gauge-reduced analytic least-squares selector). *After Corollary S22, let $y \in \mathbb{R}^{p+2M}$ collect the p free anchored damping coordinates and the real and imaginary parts of every q_m . Stack the centre residuals as*

$$\text{vec}_{\mathbb{R}} \left(q_m - \tilde{Z}_{r,m} [\bar{\lambda} + \delta\lambda_r + (i\nu_{r,m})^\beta] \right)_{r,m} = B_J y - g_J(\beta), \quad (14)$$

where all fixed contributions are included in g_J , and B_J is independent of β . Freeze positive weights in a diagonal matrix W , and suppose B_J has full column rank. To lighten notation, write $B = B_J$ and $g = g_J$ below. Then

$$\Psi_\beta(y) = \frac{1}{2} (By - g(\beta))^\top W (By - g(\beta))$$

is strictly convex, with the unique global selector

$$y_\star(\beta) = H^{-1} B^\top W g(\beta), \quad H = B^\top W B \succ 0. \quad (15)$$

The map $y_\star$ is the restriction of a real-analytic map on an open neighbourhood of I , and

$$y'_\star(\beta) = H^{-1} B^\top W g'(\beta), \quad (16)$$

$$y''_\star(\beta) = H^{-1} B^\top W g''(\beta), \quad (17)$$

where, for each positive ν ,

$$\frac{d^k}{d\beta^k} (i\nu)^\beta = (i\nu)^\beta (\log \nu + i\pi/2)^k, \quad k = 1, 2.$$

Proof. Positive diagonal W and full column rank of B give $v^\top H v = \|Bv\|_W^2 > 0$ for every $v \neq 0$. Thus the normal equation $Hy = B^\top W g(\beta)$ has the unique solution (15), which is the unique global minimizer. The principal-branch formula is analytic for $\nu > 0$; B, W, H^{-1} are constant, so differentiation gives (16)–(17). $\square$

Corollary S24 (Explicit selector-conditioning bound). *Let $\mu = \lambda_{\min}(H) > 0$, $\sigma = \|B^\top W\|_2$, and*

$$L_{r,m} = \sqrt{(\log \nu_{r,m})^2 + \pi^2/4}.$$

For $k = 1, 2$ and every $\beta \in I$,

$$\|y_\star^{(k)}(\beta)\|_2 \leq \frac{\sigma}{\mu} \left(\sum_{r,m} |\tilde{Z}_{r,m}|^2 \nu_{r,m}^{2\beta} L_{r,m}^{2k} \right)^{1/2}. \quad (18)$$

The supremum over a closed interval $A = [a, b]$ has the explicit upper bound obtained by replacing $\nu^{2\beta}$ with ν^{2b} when $\nu \geq 1$ and with ν^{2a} when $0 < \nu < 1$.

Proof. Since H is symmetric positive definite, $\|H^{-1}\|_2 = 1/\mu$. Equations (16)–(17) and submultiplicativity give $\|y_\star^{(k)}\|_2 \leq (\sigma/\mu) \|g^{(k)}\|_2$. The squared Euclidean norm of the real–imaginary pair generated by one complex derivative is $|\tilde{Z}_{r,m}|^2 \nu_{r,m}^{2\beta} L_{r,m}^{2k}$. Summation proves (18); monotonicity of $\nu^{2\beta}$ gives the endpoint rule. $\square$

Proposition S25 (Exact rational spectral-margin certificate). *In the implemented unweighted case $W = I$, suppose the exact rational Gram matrix $H = B^\top B$ is nonsingular and let $C = H^{-1}$. Define*

$$s_C = \|C\|_\infty, \quad \mu_0 = s_C^{-1}, \quad h_\infty = \|H\|_\infty.$$

Then

$$0 < \mu_0 \leq \lambda_{\min}(H), \quad \frac{\sigma^2}{\lambda_{\min}(H)^2} \leq \frac{h_\infty}{\mu_0^2}, \quad \sigma = \|B^\top\|_2. \quad (19)$$

Moreover, let $0 < \hat{\mu} \in \mathbb{Q}$ and suppose exact rational elimination gives

$$H - \hat{\mu}I = LDL^\top, \quad L_{ii} = 1, \quad D = \text{diag}(d_i), \quad d_i > 0.$$

Then

$$\hat{\mu} < \lambda_{\min}(H), \quad \frac{\sigma^2}{\lambda_{\min}(H)^2} < \frac{h_\infty}{\hat{\mu}^2}.$$

Thus a bounded rational bisection may sharpen the guaranteed baseline μ_0 ; exhaustion of the refinement budget may fall back to μ_0 without losing certification.

Proof. The matrix C is positive definite, so

$$\lambda_{\max}(C) = \lambda_{\min}(H)^{-1} \leq \|C\|_\infty = s_C.$$

This proves the first baseline inequality. Since $W = I$, $\sigma^2 = \lambda_{\max}(BB^\top) = \lambda_{\max}(B^\top B)$ on the nonzero spectrum, and $\lambda_{\max}(H) \leq \|H\|_\infty$, proving the second. For the refined claim, every nonzero x satisfies

$$x^\top (H - \hat{\mu}I)x = (L^\top x)^\top D(L^\top x) > 0.$$

Hence $H - \hat{\mu}I \succ 0$, which is equivalent to the asserted strict eigenvalue inequality. No floating-point eigensolver enters either certificate. $\square$

Remark S26 (Rate separation versus certification). For the fully free damping case, Lemma S21 is the structural threshold, while μ in Corollary S24 is its numerical margin. As the response array approaches the real-separable form (13), continuity forces $\mu \rightarrow 0$, the selector derivatives can grow, and a fixed-width order tube becomes harder to certify. This is a conditioning statement, not an information-theoretic lower bound. Moreover, μ and $\kappa_2(H)$ depend on the frozen units and column scaling of the mixed damping/morphology coordinates; they are certification margins in those coordinates, not intrinsic physical identifiability metrics. Cross-dataset comparisons require a preregistered nondimensional column scaling, after which all spectral constants must be recomputed.

Proposition S27 (Quadratic upper collapse bound near real-separable arrays). *Use the fully free physical damping coordinates and the standard real embedding of Lemma S21. Let*

$$\tilde{Z}_{r,m}^0 = \kappa_r \tilde{Z}_{r_0,m}^0, \quad \kappa_r \in \mathbb{R} \setminus \{0\},$$

and let $\tilde{Z}_{r,m} = \tilde{Z}_{r,m}^0 + \Delta Z_{r,m}$. For $B = B_{\text{full}}(\tilde{Z})$, $H = B^\top W B$, and $E_2 = \sum_{r,m} |\Delta Z_{r,m}|^2$,

$$\lambda_{\min}(H) \leq \|W\|_2 E_2.$$

In the unweighted standard embedding, if $H \succ 0$ and $E_2 > 0$, then

$$\kappa_2(H) \geq \frac{R}{E_2}.$$

If H is singular or $E_2 = 0$, the latter bound is interpreted as $+\infty$. These constants change under anchored or rescaled coordinates.

Proof. The separable matrix B_0 has a nonzero null vector by Lemma S21. For a unit null vector v_0 , with $E = B - B_0$,

$$\lambda_{\min}(H) \leq v_0^\top H v_0 = \|W^{1/2} E v_0\|_2^2 \leq \|W\|_2 \|E\|_F^2 = \|W\|_2 E_2.$$

For $W = I$, every real or imaginary q_m column has squared norm R ; hence $\lambda_{\max}(H) \geq R$, and division by the first bound proves the condition-number inequality. $\square$

Proposition S28 (Exact-Gram Krawczyk specialization). *Suppose the stored response centres and frozen weights are interpreted as exact rational numbers, the rank test passes, and $C = H^{-1}$ is computed in exact rational arithmetic. Let $\mathbf{h}(A)$ be a rigorous interval extension of $B^\top Wg(A)$. For any point y^0 and search box $\mathbf{X}$, the parameterized Krawczyk image of the normal equation is*

$$\begin{aligned}\mathbf{K} &= y^0 - C(Hy^0 - \mathbf{h}(A)) + (I - CH)(\mathbf{X} - y^0) \\ &= C\mathbf{h}(A).\end{aligned}$$

Thus $I - CH = 0$ exactly, the fixed-point contraction factor in the y -direction is zero (this is not contraction in the parameter direction), and $y_\star(\beta) \in \mathbf{K}$ for every $\beta \in A$. A replay checker can verify the exact matrix identities before evaluating the original inequalities; no approximate inverse is trusted.

Proof. Exact inversion gives $CH = I$. Substitution cancels both y^0 terms and the search-box term. Since $h(\beta) = B^\top Wg(\beta) \in \mathbf{h}(A)$, (15) gives $y_\star(\beta) = Ch(\beta) \in C\mathbf{h}(A)$. Equivalently, the fixed-point map $y \mapsto y - C(Hy - h(\beta))$ is the constant map $Ch(\beta)$. $\square$

Corollary S29 (Quadratic-width predictor tube). *Let $A = [\beta_0 - h, \beta_0 + h] \subset (0, 1]$, and let $M_j \geq \sup_{\beta \in A} |(y''_\star(\beta))_j|$. Then every selector component satisfies*

$$(y_\star(\beta))_j \in (y_\star(\beta_0))_j + (y'_\star(\beta_0))_j(\beta - \beta_0) + [-M_j h^2/2, M_j h^2/2].$$

Thus an affine predictor has a rigorously bounded $O(h^2)$ remainder. Direct interval evaluation of (15) is also valid but may be wider.

Proof. Apply Taylor's theorem with integral remainder componentwise and use $|\beta - \beta_0| \leq h$. $\square$

Remark S30 (Replayable second-order path enclosure). The machine certificate applies Corollary S29 both to $y_\star(\beta) = H^{-1}B^\top g(\beta)$ and to the correlation-preserving residual

$$r_\star(\beta) = (BH^{-1}B^\top - I)g(\beta).$$

Arb encloses the value and first derivative at the interval midpoint and the second derivative over the complete interval. Every component remainder is rounded outward to an exact dyadic rational. The final image is the intersection of this Taylor enclosure and direct interval evaluation; since the analytic path lies in both, the intersection is sound and can only be tighter. Replay reconstructs both source enclosures, the exact intersection, and every original disk and annulus inequality. The current implementation uses $W = I$; the theorem for a general frozen positive rational W is not yet an implementation claim.

Definition S31 (Full and q -eliminated feasible sets). Let Θ be the declared compact anchored damping domain, including positive physical-rate damping constraints. Define

$$\widehat{\mathcal{F}} = \left\{ (\beta, \theta, (q_m)_m) : \begin{array}{l} \beta \in I, \theta \in \Theta, q_m \in \mathcal{E}_m, \\ |q_m - d_{r,m}^{\text{dr}}(\beta, \theta) \tilde{Z}_{r,m}| \leq |d_{r,m}^{\text{dr}}(\beta, \theta)| \epsilon_{r,m} + \rho_{r,m}^q \quad \forall r, m \end{array} \right\},$$

and let

$$\mathcal{F} = \text{proj}_{(\beta, \theta)} \widehat{\mathcal{F}}.$$

Thus $\mathcal{F} \subseteq I \times \Theta$ is the exact parameter set after existentially eliminating q . The branch tree below covers this same domain $I \times \Theta$, while a selector tube constructs points of $\widehat{\mathcal{F}}$.

Theorem S32 (Certified analytic-selector inner tube). *Let $A \subset I$ be a closed order interval, and let the box $\mathbf{Y}(A)$ rigorously contain $y_\star(\beta)$ for every $\beta \in A$, using either (15) or Corollary S29. Let $\Theta_\star(A)$ be its projection onto the damping coordinates, with fixed coordinates restored. Suppose ball arithmetic proves the complete physical inclusion $\Theta_\star(A) \subseteq \Theta$, including all coupled rate-damping positivity constraints. Write $q_{\star,m}(\beta)$ and $d_{r,m}^\star(\beta)$ for the corresponding components and denominator along the analytic selector path. Suppose, additionally, that rigorous inclusion evaluations of these path expressions prove, for every r, m ,*

$$\sup_{\beta \in A} |q_{\star,m}(\beta) - d_{r,m}^\star(\beta) \tilde{Z}_{r,m}| \leq \inf_{\beta \in A} (|d_{r,m}^\star(\beta)| \epsilon_{r,m} + \rho_{r,m}^q), \quad (20)$$

$$q_{\min,m} \leq \inf_{\beta \in A} |q_{\star,m}(\beta)| \leq \sup_{\beta \in A} |q_{\star,m}(\beta)| \leq q_{\max,m}. \quad (21)$$

Then

$$\boxed{A \subseteq \text{proj}_\beta \mathcal{F}}, \quad (22)$$

with the feasible spaces of Definition S31. Strict versions of every free-coordinate, disk, and annulus inequality define a robust inner relative to the declared affine face: coordinates frozen to zero-width protocol intervals are restored exactly and are not claimed to have ambient-space margin. Rigorously established nonstrict inequalities define a closed inner certificate. An unresolved sign yields no inclusion claim.

Proof. For an arbitrary $\beta \in A$, choose the unique selector $y_\star(\beta)$. It belongs to $\mathbf{Y}(A)$. The coordinate inclusions give admissible anchored dampings, while (20) gives membership of the selected q_m in every rate-indexed forward disk and (21) gives membership in $\mathcal{E}_m$. Hence the fibre of $\tilde{\mathcal{F}}$ above every $\beta \in A$ is nonempty, which is exactly (22). Strict inequalities additionally give the reported positive margins. $\square$

Remark S33 (Correlation-preserving evaluation). The path expressions in Theorem S32 need not be evaluated over the full Cartesian product $A \times \mathbf{Y}(A)$. For the central least-squares selector, the residual is exactly

$$r_\star(\beta) = (BH^{-1}B^\top W - I)g(\beta),$$

so evaluating this expression directly preserves useful dependence on β while remaining a rigorous enclosure. Replacing it by an independent box evaluation over $\mathbf{Y}(A)$ is also sound but can be strictly wider.

Remark S34 (Selector semantics and failure). The uniqueness in Theorem S23 concerns only the frozen least-squares selector. It implies neither uniqueness of a physically feasible nuisance vector nor uniqueness of the fractional order. If the Gram matrix is singular, the selector leaves its prior box, or one inequality in Theorem S32 is unresolved, the tube returns unknown. In particular, failure of this sufficient construction is not an outer-exclusion certificate. Equating q_m to an arbitrarily selected disk centre would be unsound as a general selector: two disks can overlap in a lens containing neither centre.

Theorem S35 (Parameter-box disk hull and core). *Let B be a compact box in the anchored certification coordinates and write*

$$c_{r,m}(x) = d_{r,m}(x) \tilde{Z}_{r,m}, \quad s_{r,m}(x) = |d_{r,m}(x)| \epsilon_{r,m} + \rho_{r,m}^q.$$

Choose $x^0 \in B$ and define the exact mathematical reference $c_{r,m}^0 = c_{r,m}(x^0)$; it need not be machine-representable. Suppose rigorous ball evaluation encloses $c_{r,m}^0$ in a containing complex ball and certifies bounds

$$e_{r,m} \geq \sup_{x \in B} |c_{r,m}(x) - c_{r,m}^0|, \quad s_{r,m}^- \leq \inf_{x \in B} s_{r,m}(x), \quad s_{r,m}^+ \geq \sup_{x \in B} s_{r,m}(x).$$

Then

$$\bigcup_{x \in B} \mathcal{Q}_{r,m}(x) \subseteq \mathcal{D}(c_{r,m}^0, e_{r,m} + s_{r,m}^+) =: \mathcal{Q}_{r,m}^{\text{out}}(B), \quad (23)$$

$$\mathcal{D}(c_{r,m}^0, s_{r,m}^- - e_{r,m}) =: \mathcal{Q}_{r,m}^{\text{in}}(B) \subseteq \bigcap_{x \in B} \mathcal{Q}_{r,m}(x), \quad (24)$$

whenever $s_{r,m}^- - e_{r,m} \geq 0$ is certified. If its sign is undetermined, no inner disk may be issued.

Proof. For $q \in \mathcal{Q}_{r,m}(x)$,

$$|q - c_{r,m}^0| \leq |q - c_{r,m}(x)| + |c_{r,m}(x) - c_{r,m}^0| \leq s_{r,m}^+ + e_{r,m},$$

which proves (23). Conversely, if $|q - c_{r,m}^0| \leq s_{r,m}^- - e_{r,m}$, then for every $x \in B$,

$$|q - c_{r,m}(x)| \leq |q - c_{r,m}^0| + |c_{r,m}^0 - c_{r,m}(x)| \leq (s_{r,m}^- - e_{r,m}) + e_{r,m} = s_{r,m}^- \leq s_{r,m}(x),$$

which proves (24). In a machine proof, every comparison involving $c_{r,m}^0$ is evaluated against its entire containing ball; this can only make the test more conservative. $\square$

Proposition S36 (Finite Cartesian q -box exclusion certificate). *Fix a harmonic m . Let $C_{r,m} \subset \mathbb{C}$ be a containing complex center ball and let $R_{r,m}$ be an upper bound for the radius of $\mathcal{Q}_{r,m}^{\text{out}}(B)$. Suppose a finite exact partition of $[-q_{\max,m}, q_{\max,m}]^2$ is constructed, and every Cartesian cell P satisfies at least one strictly certified condition*

$$\inf_{q \in P} |q| > q_{\max,m}, \quad \sup_{q \in P} |q| < q_{\min,m}, \quad \text{or} \quad \inf_{\substack{q \in P \\ c \in C_{r,m}}} |q - c| > R_{r,m} \quad \text{for some } r.$$

Then

$$\left(\bigcap_r \mathcal{Q}_{r,m}^{\text{out}}(B) \right) \cap \mathcal{E}_m = \emptyset.$$

Proof. The square contains the complete annulus. By hypothesis, every point of every partition cell is either outside the closed annulus or outside at least one outer disk, so no point can belong to their joint intersection. Strict inequalities preserve all disk and annulus boundary points; an unresolved cell therefore cannot be discarded. $\square$

Theorem S37 (Certified projection sandwich). *Let a finite branch tree cover the initial compact domain without gaps and classify each leaf B as follows:*

- O : for some m , the outer disks in (23) have certified empty intersection with $\mathcal{E}_m$;
- I : for every m , one rigorously verified witness lies in all inner disks in (24) and in $\mathcal{E}_m$;
- U : neither statement has been proved.

Let $\mathcal{T}$ be any collection of order intervals certified by Theorem S32. With the exact q -eliminated parameter set $\mathcal{F}$ from Definition S31, let

$$\mathcal{A}_{\text{in}} = \left(\bigcup_{B \in I} \text{proj}_\beta B \right) \cup \left(\bigcup_{A \in \mathcal{T}} A \right), \quad \mathcal{A}_{\text{out}} = \bigcup_{B \in I \cup U} \text{proj}_\beta B,$$

where every $A \in \mathcal{T}$ lies in the initial order domain, then

$$\boxed{\mathcal{A}_{\text{in}} \subseteq \text{proj}_\beta \mathcal{F} \subseteq \mathcal{A}_{\text{out}}}. \quad (25)$$

Only when $U = \emptyset$ may the computed projection be called exact.

Proof. Theorem S35 makes every O leaf disjoint from $\mathcal{F}$ and every I leaf a subset of $\mathcal{F}$. Since $I \cup O \cup U$ covers the same initial $I \times \Theta$ domain from Definition S31, removing only O leaves cannot remove a feasible point, which gives the outer inclusion in (25). Projecting the I leaves and applying (22) to every member of $\mathcal{T}$ gives the inner inclusion. $\square$

Remark S38 (Finite-resource completeness). Ball arithmetic certifies every issued inequality. It does not remove dependency inflation or guarantee finite decisions at tangencies. Precision exhaustion, branch or leaf budgets, degenerate boundaries, and unsupported input domains must remain U . Positive-separation infeasible boxes are asymptotically excludable under convergent inclusion functions; finite classification of every boundary box is neither assumed nor claimed.

4 R2: finite-record data-to-disk theorem

Let t_n be samples in one rate window $[a, b]$, $w_n > 0$, $\sum_n w_n = 1$. For discrete WLS, also freeze exact integer sample indices $j_n \in [j_a, j_b)$. The endpoints span N complete detected R–R intervals. Put $T = b - a$ and $\hat{\omega}_r = 2\pi N/T$. The discrete estimator uses the exact nominal index phase

$$\hat{\phi}_n = 2\pi N \frac{j_n - j_a}{j_b - j_a};$$

it does not reconstruct that ratio through rounded binary64 seconds. Timestamp departures from the nominal clock enter the sampling-error budget. The continuous-lock-in branch instead uses $\hat{\phi}(t) = \hat{\omega}_r(t - a)$. The window is released only after its right R-peak anchor arrives; it is delayed-causal, not zero-delay. No rate window is independently rephased or rescaled. For estimator-basis harmonics $k \in \mathcal{K}_{\text{est}}$, define

$$\Psi_{n,k} = e^{ik\hat{\phi}_n}, \quad W = \text{diag}(w_n), \quad G = \Psi^* W \Psi.$$

Assume $G \succ 0$. Let $\hat{H}_m \neq 0$ abbreviate the independently calibrated $\hat{H}_{r,m}$ for the current rate/window under one fixed causal front end (identity preprocessing means $\hat{H}_m = 1$). It may vary with physical frequency and is not a free rate-specific morphology gain. Use

$$\tilde{C} = G^{-1} \Psi^* W \mathbf{y}, \quad \tilde{Z}_m = \tilde{C}_m / \hat{H}_m. \quad (26)$$

Theorem S39 (Conditional finite-sample enclosure). *If $\mathbf{y} = \Psi \text{diag}(\hat{H}_k) \mathbf{Z} + \mathbf{r}$ and an independent protocol guarantees $\|\mathbf{r}\|_W \leq \bar{r}$, then*

$$|\tilde{Z}_m - Z_m| \leq \frac{\sqrt{(G^{-1})_{mm}}}{|\hat{H}_m|} \bar{r} =: \epsilon_m. \quad (27)$$

Proof.

$$\tilde{C} - \text{diag}(\hat{H}_k) \mathbf{Z} = G^{-1} \Psi^* W \mathbf{r}.$$

The squared weighted operator norm of row m is

$$e_m^* G^{-1} \Psi^* W \Psi G^{-1} e_m = (G^{-1})_{mm}.$$

Cauchy–Schwarz followed by division by $|\hat{H}_m|$ gives (27). $\square$

To make $\bar{r}$ operational, let the record start at t_0 , let a rate segment switch at $t_{\text{switch}} \geq t_0$, set $S = (t_{\text{switch}} - t_0)/\tau_*$, and impose $z(t_0) = z_0$. Let u_{pre} act on $[t_0, t_{\text{switch}})$ and u_0 after the switch. Let the estimation window start after burn-in b_0 , and put $s_n = (t_n - t_{\text{switch}})/\tau_* \in [b_0, b_0 + T/\tau_*]$ and

$$M_{\alpha,\lambda}(s) = E_\alpha(-\lambda s^\alpha), \quad k_{\alpha,\lambda}(s) = s^{\alpha-1} E_{\alpha,\alpha}(-\lambda s^\alpha).$$

Assume the pre-switch and new nominal inputs obey $\|u_{\text{pre}}\|_{\infty} \leq U_{\text{pre}}$ and $\|u_0\|_{\infty} \leq U_0$. The finite-lower-limit and switch-memory envelopes are

$$B_{\text{hist}}(s) = \left(|z_0| + \frac{U_0}{\lambda}\right) M_{\alpha,\lambda}(S+s), \quad (28)$$

$$B_{\text{dwell}}(s) = \frac{U_{\text{pre}} + U_0}{\lambda} [M_{\alpha,\lambda}(s) - M_{\alpha,\lambda}(S+s)]. \quad (29)$$

If there is additional unmodelled pre-record Caputo history, an independently calibrated term $C_{\text{pre}} M_{\alpha,\lambda}(S+s)$ is mandatory. For a nonidentity causal front end with impulse response h , replace each time-domain envelope B by $|h| * B$ before taking its weighted norm.

Define a separate operational beatwise R-R phase by linear interpolation between detected R peaks. It is not the WLS basis above: its deviation from the fixed nominal phase is propagated as an HRV perturbation. Let the total certified phase error be $\Delta_{\phi}(s)$, including the observable warp, R-time error, and any claimed latent-phase discrepancy. If $|R_j - \hat{R}_j| \leq \Delta_R$, the detection part can be bounded within beat j by

$$\Delta_{\phi,R,j} \leq \frac{2\pi\Delta_R}{\hat{T}_j - 2\Delta_R}, \quad \hat{T}_j > 2\Delta_R.$$

With $d(x) = 2\sin(\min\{x, \pi\}/2)$ and independently bounded input harmonics $|q_k| \leq \bar{Q}_k$, require either a finite forcing basis or $\sum_k \bar{Q}_k < \infty$, and put

$$\Delta U(s) = \sum_{k \in \mathbb{Z}} \bar{Q}_k d(|k| \Delta_{\phi}(s)), \quad (30)$$

$$B_{\text{HRV}}(s) = \int_0^s k_{\alpha,\lambda}(s-v) \Delta U(v) dv. \quad (31)$$

When $\Delta U \leq \bar{\Delta U}$, complete monotonicity gives

$$B_{\text{HRV}}(s) \leq \frac{\bar{\Delta U}}{\lambda} [1 - M_{\alpha,\lambda}(s)].$$

Thus HRV is propagated through the fractional dynamics; it is not merely an output-phasor rotation.

Any remaining rate-specific anchor mismatch that is not already included in Δ_{ϕ} must have an independently certified output-residual envelope $B_{\text{phase}}(s)$, with disjoint provenance and

$$\bar{r}_{\text{phase}} \geq \left(\sum_n w_n B_{\text{phase}}(s_n)^2 \right)^{1/2}.$$

For the discrete WLS estimator, define

$$a_m = \sqrt{(G^{-1})_{mm}}, \quad h_{mk} = e_m^* G^{-1} \Psi^* W \psi_k.$$

A sufficient coefficient-level radius is

$$\begin{aligned} \epsilon_{r,m} = & \frac{a_m}{|\hat{H}_m|} (\bar{r}_{\text{hist}} + \bar{r}_{\text{dwell}} + \bar{r}_{\text{HRV}} + \bar{r}_{\text{phase}} + \bar{r}_{\text{sample}} + \bar{r}_{\text{model}} + \bar{r}_{\text{noise}}) \\ & + \epsilon_{r,m}^{\text{omitted}} + \epsilon_{r,m}^{\text{delay}} + \epsilon_{r,m}^{\text{quad}} + \epsilon_{r,m}^{\text{filt-init}}, \end{aligned} \quad (32)$$

where, for $j \in \{\text{hist}, \text{dwell}, \text{HRV}\}$,

$$\bar{r}_j \geq \sup_{(\alpha,\lambda) \in I \times \Lambda} \left(\sum_n w_n B_j(s_n)^2 \right)^{1/2}. \quad (33)$$

The remaining terms require the following independent bounds.

Window leakage and harmonic tail. If $\sum_k \overline{|H_{r,k}|} \overline{Z}_{r,k} < \infty$, then

$$\epsilon_{r,m}^{\text{omitted}} \leq \sum_{k \notin \mathcal{K}_{\text{est}}} \frac{|h_{mk}| \overline{|H_{r,k}|}}{|\widehat{H}_m|} \overline{Z}_{r,k}.$$

For a rectangular single-harmonic window of phase length $L = \widehat{\omega}_r T$,

$$h_{mk} = e^{i(k-m)L/2} \text{sinc}((k-m)L/2).$$

The nominal leakage vanishes when $L = 2\pi N$, but transient, HRV, and model errors do not.

Sampling and interpolation. For sample-time error Δt_n , ADC step Δ_{ADC} , and independently bounded interpolation error,

$$\bar{r}_{\text{sample}} \geq \left[\sum_n w_n (M_1 \Delta t_n + \Delta_{\text{ADC}}/2 + \epsilon_{\text{interp},n} + \epsilon_{\text{aa},n})^2 \right]^{1/2}.$$

If a continuous lock-in integral is instead approximated by the trapezoidal rule, freeze the decomposition

$$y(t_n) = x_{\text{nom}}^{\text{ret}}(t_n) + e_n.$$

Here $x_{\text{nom}}^{\text{ret}}$ is exactly the retained-harmonic, nominal-front- end reconstruction whose continuous coefficient defines the target; every dynamic, acquisition, noise, and model term handled by samplewise bounds is placed in e_n . Bound the trapezoidal functional of e_n by its declared sample envelopes, and add the following quadrature remainder only for the nominal reconstruction:

$$\epsilon_{r,m}^{\text{quad}} \leq \frac{1}{12T|\widehat{H}_m|} \sum_j \Delta t_j^3 M_{2,j}, \quad M_{2,j} \geq \sup_{[t_j, t_{j+1}]} |(x_{\text{nom}}^{\text{ret}})'' - 2im\widehat{\omega}_r (x_{\text{nom}}^{\text{ret}})' - (m\widehat{\omega}_r)^2 x_{\text{nom}}^{\text{ret}}|.$$

Using the second derivative of the actual signal here would mix two incompatible decompositions: its continuous residual is not controlled by sample values alone. The constants $M_1, M_{2,j}$ cannot be certified from the same samples without an anti-aliasing, bandwidth, derivative, or interval-reconstruction assumption.

Front-end delay, noise, and model discrepancy. If

$$\frac{H_{r,m}}{\widehat{H}_m} = (1 + \delta_a) e^{-i\delta_\psi}, \quad |\delta_a| \leq a_H, \quad |\delta_\psi| \leq |m|\widehat{\omega}_r \Delta_\tau + \Delta_{\psi,0},$$

then

$$\epsilon_{r,m}^{\text{delay}} \leq \overline{Z}_{r,m} [a_H + (1 + a_H) d(|m|\widehat{\omega}_r \Delta_\tau + \Delta_{\psi,0})].$$

A common phase angle may be absorbed into q_m ; a common delay in seconds may not, because its phase depends on heart rate. Deterministic noise obeys

$$\|\mathbf{v}\|_W \leq \bar{v}_W \implies \bar{r}_{\text{noise}} := \bar{v}_W,$$

for discrete WLS; this discrete norm needs its own calibration. A continuous L_2 bound alone cannot control point samples without an independently certified sampling or reconstruction inequality. For the exact continuous lock-in functional, however, Cauchy–Schwarz gives directly

$$\left| \frac{1}{T} \int_a^b v(t) e^{-im\widehat{\omega}_r t} dt \right| \leq \frac{N_2}{\sqrt{T}}, \quad \|v\|_{L_2(a,b)} \leq N_2.$$

The trapezoidal implementation then also needs the separately certified quadrature contract stated above. A Gaussian or RMS SNR gives a confidence disk, not a deterministic disk. If the

causal filter starts at t_0 , its pre-start input is bounded by $Y_{\max}$, and its impulse response is h , define the initialization envelope

$$B_{\text{init}}(t) := Y_{\max} \int_{t-t_0}^{\infty} |h(s)| ds.$$

The extraction-functional contract must then be respected. For exact continuous lock-in,

$$\epsilon_{r,m}^{\text{filt-init,cont}} \leq \frac{1}{T|\widehat{H}_m|} \int_a^b B_{\text{init}}(t) dt.$$

For discrete WLS, with the row norm a_m of Theorem S39,

$$\epsilon_{r,m}^{\text{filt-init,WLS}} \leq \frac{a_m}{|\widehat{H}_m|} \left(\sum_n w_n B_{\text{init}}(t_n)^2 \right)^{1/2}.$$

For the trapezoidal sample functional with internally derived weights ω_n^{trap} ,

$$\epsilon_{r,m}^{\text{filt-init,trap}} \leq \frac{1}{T|\widehat{H}_m|} \sum_n \omega_n^{\text{trap}} B_{\text{init}}(t_n).$$

The last bound concerns the sampled initialization residual; any continuous quadrature remainder remains governed by the separately frozen decomposition above. The model term $\bar{r}_{\text{model}}$ must likewise be bounded independently; choosing it from the residual of the same fitted surrogate is circular.

Theorem S40 (Conditional finite-record data-to-disk map). *Under the dimensionless model, delayed-causal phase protocol, calibrated front end, and every independent deterministic bound in (28)–(32),*

$$|\widetilde{Z}_{r,m} - Z_{r,m}| \leq \epsilon_{r,m}.$$

Consequently the response disk in (9) has guaranteed coverage. The sum is a valid outer disk; it need not be the smallest disk when errors are correlated.

Proof. Equations (28) and (29) follow by splitting the scalar Mittag–Leffler convolution at the recording and rate-switch boundaries. Equation (31) is the same convolution applied to the bounded input perturbation (30). The remaining displayed inequalities bound linear extraction, leakage, sampling, calibration, noise, and initialization terms. The triangle inequality and Theorem S39 give (32). $\square$

Proposition S41 (Elementary uniform Mittag–Leffler envelope). *Let $I = [\alpha_-, \alpha_+] \subset (0, 1]$, $\Lambda = [\lambda_-, \lambda_+] \subset \mathbb{R}_{>0}$, and*

$$p_I(t) = \begin{cases} 0, & t = 0, \\ t^{\alpha_+}, & 0 < t \leq 1, \\ t^{\alpha_-}, & t \geq 1, \end{cases} \quad R_I(t) = \frac{1}{1 + \lambda_- p_I(t)}.$$

Then, uniformly for $(\alpha, \lambda) \in I \times \Lambda$ and $t \geq 0$,

$$0 < E_{\alpha}(-\lambda t^{\alpha}) \leq R_I(t). \quad (34)$$

Consequently,

$$B_{\text{hist}}(s) \leq \left(|z_0| + \frac{U_0}{\lambda_-} + C_{\text{pre}} \right) R_I(S + s), \quad (35)$$

$$B_{\text{dwell}}(s) \leq \frac{U_{\text{pre}} + U_0}{\lambda_-} R_I(s). \quad (36)$$

Proof. For $0 < \alpha < 1$, Simon's uniform bound gives

$$E_\alpha(-x) \leq \frac{1}{1 + x/\Gamma(1 + \alpha)}, \quad x \geq 0$$

[22]. Log-convexity of Γ , together with $\Gamma(1) = \Gamma(2) = 1$, gives $\Gamma(1 + \alpha) \leq 1$, hence $E_\alpha(-x) \leq (1 + x)^{-1}$. At $\alpha = 1$, $E_1(-x) = e^{-x} \leq (1 + x)^{-1}$. By construction, $p_I(t) \leq t^\alpha$ and $\lambda_- \leq \lambda$, proving the upper bound. Pollard's complete monotonicity theorem gives positivity and monotone decrease of $E_\alpha(-x)$ [21]. Applying these facts to (28)–(29), and dropping the nonnegative subtracted dwell term, proves the two envelopes. $\square$

Proposition S42 (Replayable WLS Gram margin). *Suppose complex balls G_{jk} contain every entry of the Hermitian WLS Gram matrix. Let $\underline{g}_{jj}$ be a lower bound for its real diagonal, $\bar{g}_{jj}$ an upper bound, and $\bar{a}_{jk} \geq |G_{jk}|$ for $j \neq k$. Define*

$$g_- = \min_j \left(\underline{g}_{jj} - \sum_{k \neq j} \bar{a}_{jk} \right), \quad g_+ = \max_j \left(\bar{g}_{jj} + \sum_{k \neq j} \bar{a}_{jk} \right).$$

If $g_- > 0$, then

$$G \succ 0, \quad \lambda_{\min}(G) \geq g_-, \quad \lambda_{\max}(G) \leq g_+, \quad \kappa_2(G) \leq \frac{g_+}{g_-},$$

and every WLS row factor satisfies

$$\sqrt{(G^{-1})_{mm}} \leq g_-^{-1/2}.$$

If $g_- > 0$ cannot be proved, this test is inconclusive rather than a singularity certificate.

Proof. The first two spectral inclusions are the Hermitian Gershgorin theorem applied with outward-containing entries. Positivity of every lower endpoint implies positive definiteness. Spectral inversion then gives $(G^{-1})_{mm} \leq \lambda_{\max}(G^{-1}) = 1/\lambda_{\min}(G)$. $\square$

Proposition S43 (Verified shifted Hermitian LDL* fallback). *Let $\mathbf{G}_{jk}$, $j \geq k$, be lower-triangular complex balls containing the corresponding entries of one exact Hermitian Gram matrix G ; the upper triangle is reconstructed by conjugate symmetry. Let $\mu > 0$ be rational and apply unpivoted ball LDL* elimination to $\mathbf{G} - \mu I$. At each step retain the real projection of the pivot only after its imaginary ball is verified to contain zero. If every real pivot ball has a strictly positive lower endpoint, then*

$$G - \mu I \succ 0, \quad \lambda_{\min}(G) > \mu.$$

Together with the Gershgorin upper bound g_+ from Proposition S42, this gives

$$\kappa_2(G) < \frac{g_+}{\mu}, \quad \sqrt{(G^{-1})_{mm}} < \mu^{-1/2}.$$

Failure of any pivot sign is inconclusive.

Proof. Inductively, ball operations contain the unique exact unpivoted LDL* factors of the represented Hermitian matrix. Every exact Schur-complement pivot is real; the imaginary containment check guards that invariant before the real projection is used. Strict positivity of all exact pivots proves $G - \mu I \succ 0$ by the Hermitian Sylvester criterion. The spectral bounds then follow as in Proposition S42. $\square$

Proposition S44 (Exact-integer-payload WLS numerical enclosure). *Freeze a selected digital payload $d_n \in \mathbb{Z}$, $n = 1, \dots, n_s$, integer baseline b_{ADC} , positive binary-rational gain g_{ADC} , integer sample indices j_n , window endpoints $a < b$, complete RR count N , and the contiguous two-sided basis $\mathcal{K}_{\text{est}}$. Interpret*

$$y_n = \frac{d_n - b_{\text{ADC}}}{g_{\text{ADC}}}, \quad w_n = \frac{1}{n_s}, \quad \phi_n = 2\pi N \frac{j_n - a}{b - a}$$

exactly. Suppose complex ball arithmetic directly encloses every $e^{ik\phi_n}$, accumulates containing balls $\mathbf{G} \ni \Psi^ W \Psi$ and $\mathbf{b} \ni \Psi^* W y$, and a verified linear solve returns $\mathbf{C}$ containing the exact solution. Let the frozen front-end value $\hat{H}_m \neq 0$ be interpreted as an exact binary complex rational, and let*

$$\mathbf{Z}_m^{\text{func}} = \mathbf{C}_m / \hat{H}_m, \quad \rho_m \geq \sup\{|z - \tilde{Z}_m| : z \in \mathbf{Z}_m^{\text{func}}\}$$

be an outward dyadic bound. Then the exact mathematical sample-functional coefficient satisfies

$$|\mathbf{Z}_m^{\text{func}} - \tilde{Z}_m| \leq \rho_m.$$

Consequently ρ_m is a sufficient numerical floor for the sampling radius of a disk centred at the stored binary64 $\tilde{Z}_m$.

Proof. The integer and binary-rational inputs are exact. Inclusion isotonicity of Arb/ACB operations [19] contains every exponential, finite sum, and product in the displayed exact G and b . A successful verified solve contains the unique exact C ; division by the nonzero exact $\hat{H}_m$ contains $\mathbf{Z}_m^{\text{func}}$. The final complex-ball modulus and outward dyadic comparison give the asserted disk inequality. $\square$

Remark S45 (Scope of the exact-payload enclosure). Proposition S44 closes phase evaluation, G, b accumulation, linear-solve, and stored-centre rounding for the frozen sample functional. It does not bound sensor error, ADC-gain calibration uncertainty, R-annotation error, subsampling aliasing, front-end calibration uncertainty, or model discrepancy. Those remain independent R2 terms. In particular, $\mathbf{Z}_m^{\text{func}}$ is not the true model response Z_m ; coverage of the latter requires the complete remaining R2 budget.

Theorem S46 (Machine-certified partial R2 component map). *Fix exact binary-rational representations of the sample times, protocol intervals, and independently sourced primitive bounds. Normalize declared positive weights exactly for discrete WLS; for continuous lock-in, derive the unique nonuniform composite-trapezoid weights internally from those exact timestamps. For the dynamic residual, require either an exactly identity front end or independently certified per-sample output envelopes containing $(|h| * B_{\text{hist}})(s_n)$ and $(|h| * B_{\text{dwell}})(s_n)$. For estimator extraction, use either the discrete WLS map of (26), or a complete-cycle continuous lock-in approximated with the exact nonuniform composite-trapezoid weights; in the latter case the continuous harmonic Gram is I , and the panelwise quadrature remainder is added separately with decomposition target $x_{\text{nom}}^{\text{ret}}$ frozen as above.*

Suppose ball arithmetic proves either Proposition S42 or Proposition S43 for the discrete path (or verifies the exact $G = I$ continuous path) and evaluates the applicable dynamic output envelopes, sampling norm, quadrature remainder, and front-end delay expression with outward rounding. Then the resulting history, dwell, sampling, quadrature, delay, Gram, condition-number, and extraction-amplification upper bounds are simultaneous deterministic component bounds for the frozen window.

A certificate that binds the complete input manifest, evidence hashes, backend and precision, outward rational enclosures, and a canonical digest can be independently replayed by reconstructing every derived quantity. Failure of a source contract, a positive Gram lower bound, ball comparison, or a resource limit has only UNKNOWN/NOT_CERTIFIABLE semantics.

Proof. Proposition S41 bounds the two model-input memory sequences pointwise. Identity propagation preserves them; otherwise the separately certified convolved envelopes are used. Positive normalized weights and monotonicity of the square root preserve their weighted-norm bounds. Propositions S42 and S43 supply the discrete-WLS coefficient amplification. Orthogonality over a complete integer number of cycles gives $G = I$ for the continuous lock-in path, while requiring the exact composite weights binds its sampled implementation. The sample bound follows from the pointwise triangle inequality, and the trapezoid term follows panelwise from the Peano-kernel error bound. Finally, $|e^{-i\theta} - 1| = 2|\sin(\theta/2)|$, capped by 2, gives the declared delay expression. All operations are inclusion isotone under outward ball arithmetic. Recalculation from the bound input rather than trust in stored derived fields proves replay. $\square$

Theorem S47 (Frozen-selector R2-to-T1 margin transfer). *Let one T1 certificate provide a selector path $(d_{r,m}^*(\alpha), q_m^*(\alpha))$ on a closed interval A and certified old disk margins*

$$\mu_{r,m}(\alpha) = |d_{r,m}^*| \epsilon_{r,m} + \rho_{r,m}^q - |q_m^* - d_{r,m}^* \tilde{Z}_{r,m}| \geq \underline{\mu}_{r,m} \geq 0.$$

Keep this old path fixed. Consider an independently certified family of new R2 disks

$$\tilde{Z}' = \tilde{Z} + \xi, \quad \epsilon' = \epsilon + a, \quad \rho^{q'} = \rho^q + b,$$

with nonnegative new radii. If $0 < d^- \leq |d^| \leq d^+$, $|\xi| \leq \zeta$, $a \geq \underline{a}$, and $b \geq \underline{b}$, define*

$$c = \zeta - \underline{a}, \quad L^q = \begin{cases} d^+ c - \underline{b}, & c \geq 0, \\ d^- c - \underline{b}, & c < 0. \end{cases}$$

Then every member of the new family has, along the same path,

$$\mu'_{r,m}(\alpha) \geq \underline{\mu}_{r,m} - L^q_{r,m}. \quad (37)$$

If all transferred disk, coordinate, damping-facet, and annulus margins are strictly positive, A remains a robust inner interval for every member of the family. If all are nonnegative, it remains a closed inner interval. Otherwise this transfer is unknown and makes no outer-exclusion claim.

Proof. The triangle inequality gives

$$|q^* - d^*(\tilde{Z} + \xi)| \leq |q^* - d^* \tilde{Z}| + |d^*| |\xi|.$$

After adding the new radius,

$$\mu' \geq \mu - |d^*|(\zeta - \underline{a}) + \underline{b}.$$

Maximizing the affine expression $|d^*|(\zeta - \underline{a}) - \underline{b}$ over $[d^-, d^+]$ selects the displayed endpoint according to the sign of c , which proves (37). The old path already satisfies the model equations and all unchanged priors. Transferred nonnegative margins therefore supply one witness for every α and every admitted R2 perturbation; strict margins give the robust version. $\square$

Remark S48 (Direction of the transfer theorem). Increasing a response or independent morphology radius while holding its centre fixed can only help inner feasibility, although it weakens exclusion power and enlarges the outer parameter set. Thus Theorem S47 is a budget for centre motion, radius shrinkage, and prior tightening; it is not by itself a statement that a larger data-error radius is higher quality. Rebuilding the least-squares selector from the new centres would prove a different path and is not a valid replay of the old witness. The current transfer certificate binds the declared perturbation family and the old T1 certificate; binding that family to a named R2 component-certificate bundle is a separate integration gate.

For probabilistic components, the coverage event must be declared simultaneously over every selected rate-harmonic disk used by R1. Unless a direct joint calibration is supplied, its failure budget includes all error terms, windows, rates, and harmonics through a preregistered union bound. An empty R1 set obtained from such disks is therefore a confidence-qualified rejection, not a deterministic contradiction.

Corollary S49 (Certified minimum burn-in). *Let ϵ_{rest} contain every radius term except history and dwell, and let $\epsilon_\star > \epsilon_{\text{rest}}$. For the identity front end, or after an independently calibrated filter-envelope factor is absorbed, with*

$$C_H(\alpha, \lambda) = |z_0| + C_{\text{pre}} + \frac{U_{\text{pre}} + 2U_0}{\lambda},$$

a sufficient dimensionless burn-in is

$$B_{\min} = \inf \left\{ B \geq 0 : \frac{a_m}{|\widehat{H}_m|} \sup_{I \times \Lambda} C_H(\alpha, \lambda) E_\alpha(-\lambda B^\alpha) \leq \epsilon_\star - \epsilon_{\text{rest}} \right\}.$$

The physical dwell to the right window endpoint must satisfy

$$T_{\text{dwell}} \geq \tau_\star B_{\min} + \frac{2\pi N}{\widehat{\omega}_r}.$$

This is a minimum for the declared certificate, not a physiological shortest settling time. The sum in C_H matches the separately budgeted history and dwell norms in (32): each is bounded by its coefficient times $E_\alpha(-\lambda B^\alpha)$. A smaller $\max\{|z_0| + U_0/\lambda + C_{\text{pre}}, (U_{\text{pre}} + U_0)/\lambda\}$ is available only if their shared $M(s) - M(S + s)$ structure is certified as one joint residual budget rather than added as two independent radii.

Corollary S50 (Minimum deterministic energy SNR). *Suppose $|Z_{r,m}| \geq \underline{Z}_{r,m} > 0$, the target relative radius is η , and $\eta \underline{Z}_{r,m} > \epsilon_{\text{rest}}$. Suppose in addition that the declared estimator has a certified bridge $\bar{r}_{\text{noise}} \leq C_s N_2 / \sqrt{T}$. Exact continuous lock-in has $C_s = 1$ and coefficient functional norm $a_m = 1$; discrete WLS requires a separate sampling/reconstruction theorem before such a bridge may be used. With*

$$\text{SNR}_{E,r,m} = \frac{|\widehat{H}_m| \underline{Z}_{r,m} \sqrt{T}}{N_2},$$

the sufficient condition

$$\text{SNR}_{E,r,m} \geq \frac{a_m C_s}{\eta - \epsilon_{\text{rest}} / \underline{Z}_{r,m}}$$

guarantees the target. Under only pointwise worst-case noise, lengthening the window need not improve this deterministic SNR.

Remark S51 (R2 is conditionally closed, but not calibrated on real ECG). The algebraic data-to-disk theorem and the machine propagation of history, dwell, sampling, optional quadrature, relative delay, and WLS Gram bounds are now closed conditional on their declared primitive envelopes. The project does not yet possess independent real-ECG numerical bounds for R-time and latent intra-beat phase error, measurement/model discrepancy, anti-aliasing or interpolation, relative filter delay, Caputo prehistory, and the harmonic tail. The component certificate also lists HRV, phase anchor, filter initialization, measurement noise, and model residual as omitted terms, so it cannot by itself issue a complete R2 pass. Without every required bound, (9) cannot be invoked on real ECG. An absent bound yields NOT_CERTIFIABLE, not model rejection.

5 R3: worst-case ambiguity and design

Let $\mathfrak{D}$ be a compact design set, Θ a compact nuisance set, and $f_{\mathcal{D}}(\alpha, \theta) \in \mathbb{C}^n$ a jointly continuous nominal output map in $(\mathcal{D}, \alpha, \theta)$, with one fixed output dimension n across designs. Let

$$\mathcal{U}_{\mathcal{D}} = \{e : |e_j| \leq \epsilon_j(\mathcal{D})\}, \quad \epsilon_j(\mathcal{D}) \geq \epsilon_{\min} > 0,$$

with continuous radii. Define

$$\mathcal{Y}_{\mathcal{D}}(\alpha) = \{f_{\mathcal{D}}(\alpha, \theta) + e : \theta \in \Theta, e \in \mathcal{U}_{\mathcal{D}}\}, \quad \mathcal{A}_{\mathcal{D}}(y) = \{\alpha : y \in \mathcal{Y}_{\mathcal{D}}(\alpha)\}.$$

Write $\mathcal{O}_{\mathcal{D}} = \bigcup_{\alpha \in I} \mathcal{Y}_{\mathcal{D}}(\alpha)$ for the reachable observation set, and define the class of uniformly reliable set estimators by

$$\mathfrak{R}_{\mathcal{D}} = \{S : \mathcal{O}_{\mathcal{D}} \rightarrow 2^I : \alpha \in S(f_{\mathcal{D}}(\alpha, \theta) + e) \quad \forall \alpha \in I, \theta \in \Theta, e \in \mathcal{U}_{\mathcal{D}}\}.$$

Theorem S52 (Minimal reliable set and ambiguity identity). *Every deterministic set estimator guaranteed to contain the true order for all admissible realizations must contain $\mathcal{A}_{\mathcal{D}}(y)$ pointwise. Moreover,*

$$\inf_{S \in \mathfrak{R}_{\mathcal{D}}} \sup_{y \in \mathcal{O}_{\mathcal{D}}} \text{diam } S(y) = \sup_{y \in \mathcal{O}_{\mathcal{D}}} \text{diam } \mathcal{A}_{\mathcal{D}}(y) = \Delta(\mathcal{D}), \quad (38)$$

where

$$\Delta(\mathcal{D}) = \max\{|\alpha - \beta| : \mathcal{Y}_{\mathcal{D}}(\alpha) \cap \mathcal{Y}_{\mathcal{D}}(\beta) \neq \emptyset\}. \quad (39)$$

Proof. Every $\alpha \in \mathcal{A}_{\mathcal{D}}(y)$ can be the true order of an admissible experiment producing y , so a uniformly reliable set must contain it. Conversely, $S(y) = \mathcal{A}_{\mathcal{D}}(y)$ is itself reliable, which proves the first equality in (38). Two orders occur in one consistency set exactly when their output sets intersect. Compactness and continuity attain the displayed maximum. $\square$

Definition S53 (Separation modulus). For $0 < \delta \leq \text{diam } I$, let

$$s_{\mathcal{D}}(\alpha, \beta) = \min_{\theta, \theta' \in \Theta} \max_j \frac{|f_{\mathcal{D},j}(\alpha, \theta) - f_{\mathcal{D},j}(\beta, \theta')|}{\epsilon_j(\mathcal{D})}, \quad (40)$$

$$\mu_{\mathcal{D}}(\delta) = \min_{\substack{\alpha, \beta \in I \\ |\alpha - \beta| \geq \delta}} s_{\mathcal{D}}(\alpha, \beta). \quad (41)$$

Theorem S54 (Exact disk threshold and optimal-design existence). *For product complex-disk errors,*

$$\mathcal{Y}_{\mathcal{D}}(\alpha) \cap \mathcal{Y}_{\mathcal{D}}(\beta) \neq \emptyset \iff s_{\mathcal{D}}(\alpha, \beta) \leq 2,$$

and hence

$$\Delta(\mathcal{D}) < \delta \iff \mu_{\mathcal{D}}(\delta) > 2.$$

For fixed δ , $\mathcal{D} \mapsto \mu_{\mathcal{D}}(\delta)$ is continuous and

$$\mathcal{D}_{\delta}^* \in \arg \max_{\mathcal{D} \in \mathfrak{D}} \mu_{\mathcal{D}}(\delta)$$

exists. If the optimum exceeds 2, a maximizing design with the canonical estimator $S(y) = \mathcal{A}_{\mathcal{D}}(y)$ guarantees worst-case diameter less than δ . If the optimum is at most 2, no uniformly reliable deterministic set estimator under any design in $\mathfrak{D}$ can guarantee diameter less than δ .

Proof. A common observation implies coordinatewise nominal-output differences at most $2\epsilon_j$. Conversely, the midpoint of two nominal outputs whose coordinate differences are at most $2\epsilon_j$ lies in both disk products. This proves the threshold. Continuity follows by successive minimization of a continuous function over fixed compact sets; maximization over compact $\mathfrak{D}$ attains an optimum. $\square$

Theorem S55 (Local close-rate condition numbers). *Fix $x, q \neq 0, \lambda > 0$, and $\alpha \in \text{int } I$. Let $\mathcal{R}_{2,h}$ denote the locally defined physical two-rate recovery map at log-rates $x, x+h$. For three equal log steps define*

$$Q_h(W) = \frac{W(x+2h) - W(x)}{W(x+h) - W(x)}, \quad \mathcal{R}_{3,h}(W) = \frac{1}{h} \log(\text{Re } Q_h(W) - 1)$$

on the open neighborhood where the denominator is nonzero, $\text{Re } Q_h > 1$, and the displayed value lies in $\text{int } I$. Thus $\mathcal{R}_{3,h}$ is a declared real-part projection diagnostic for off-model complex perturbations, not a physical inverse of an arbitrary complex quotient. For independent complex inverse-response perturbations with the ℓ_∞ norm,

$$\|D_W \mathcal{R}_{2,h}\|_{\infty \rightarrow \mathbb{R}} = \Theta(h^{-1}), \quad \|D_W \mathcal{R}_{3,h}\|_{\infty \rightarrow \mathbb{R}} = \Theta(h^{-2})$$

as $h \rightarrow 0^+$.

Proof. For two rates, $D_h = W(x+h) - W(x) = \Theta(h)$, so differentiating $C = W_1/(W_2 - W_1)$ gives $\Theta(h^{-2})$. The derivative of H_C with respect to C contains $e^{\beta h} - 1 = \Theta(h)$, while at the true order

$$\frac{\partial H_C}{\partial \beta} = \frac{e^{\alpha h} h}{e^{\alpha h} - 1} + \frac{\lambda}{\nu_1^\alpha} \frac{\pi}{2 \sin(\pi \alpha / 2)} = \Theta(1).$$

The implicit sensitivity is $\Theta(h^{-1})$. For three equal log steps,

$$F_h(\alpha) = 1 + e^{\alpha h}, \quad F'_h(\alpha) = h e^{\alpha h} = \Theta(h),$$

while $D_W Q_h = \Theta(h^{-1})$. The real-part projection has unit norm and admits a perturbation direction attaining that order; composing it with F_h^{-1} therefore gives $\Theta(h^{-2})$. $\square$

These are worst-direction condition numbers of the stated algebraic maps, not information-theoretic lower bounds for all three-rate estimators. The corresponding finite-error expansions require respectively $\epsilon = o(h)$ and $\epsilon = o(h^2)$; fixed nonzero error eventually leaves the local regime.

Proposition S56 (Finite frequencies do not identify mechanism). *Let $\{(s_j, z_j)\}_{j=1}^L$ be distinct finite frequency-response data with $s_j = i\omega_j \in i\mathbb{R}$, closed under complex conjugation with conjugate-consistent targets. For any $a > 0$ and integer $N \geq L$, there is a real-coefficient, stable, strictly proper rational transfer function*

$$H_{\text{rat}}(s) = \frac{P(s)}{(s+a)^N}$$

such that $H_{\text{rat}}(s_j) = z_j$ for every j .

Proof. Interpolate the conjugate-consistent targets $z_j(s_j + a)^N$ by the unique polynomial P of degree at most $L - 1$. Conjugate symmetry and uniqueness make its coefficients real. The denominator has only the stable pole $-a$, and $N > \deg P$. $\square$

Consequently, uniqueness inside the declared fractional family identifies an effective surrogate order, not an intrinsic fractional physiological mechanism. Held-out transient prediction against integer-order rational baselines is therefore essential.

6 Gate status and claim boundary

Gate	Current status	Blocking item
R0	pass	assumptions and compact-prior checks
R1	core, spectral/predictor tube, and transfer pass; exact closure open	unresolved leaves, sufficient selector, and finite budgets
R2	component certificate pass; real calibration open	independent and omitted bounds in Remark S51
R3	theorem pass	globally certified numerical optimizer
R4	narrow pass	no claim of generic set/design novelty

Result semantics.

R2 returns one of

PASS_DETERMINISTIC, PASS_PROBABILISTIC,
NOT_CERTIFIABLE, EXCLUDE_PROTOCOL.

Downstream REJECT_MODEL requires R2 disks with declared simultaneous coverage and an R1 proof that the entire preregistered joint domain is empty, with no unknown leaf. A probabilistic disk yields only a confidence-qualified rejection. Missing protocol bounds, singular certificates, unresolved global boxes, and nonempty but overly wide order sets remain NOT_CERTIFIABLE.

Excluded claim. The theory outputs an order set. It does not prove stability of an observer implemented with a mismatched fractional derivative; that problem contains a full-history operator mismatch and is deferred.

Prior-art boundary. Bounded complex-frequency set inversion for an isomorphic fractional first-order model and complex-disk set estimation already exist [3, 4], while frequency-domain fractional identifiability and its practical sensitivity limits are established topics [5]. Fractional Kalman filters already estimate states, parameters, and order, and recent high-order FO-EKF variants further reinforce that *FO-EKF* itself is not a novelty claim [1, 2]. Robust fractional experiment design and generic set-membership design also exist [8, 9]. Finite-memory reset/truncation observers are prior art [10, 11]; fractional ECG generators and ECG-calibrated fractional McSharry models already exist [12, 13]; and contemporary cardiac twins expose serious surface-ECG nonuniqueness [14, 15]. Parameter-dependent interval verification, affine sensitivity enclosures, existential projection, and validated solution tubes are likewise established methods [17, 16, 18, 20]. We therefore do not propose a new interval Newton or validated-continuation method. The defensible contribution is restricted to their FO-ECG specialization: a dimensionless two-rate quotient theorem modulo one common complex morphology/lead gauge, the rank characterization in Lemma S21, the analytic witness tube paired with the outer disk cover, the frozen-witness finite-record margin transfer, the joint multi-rate consistency semantics, and the finite-record uncertainty/design chain. We do not claim priority with the word first.

References

- [1] D. Sierociuk and A. Dzieliński, Fractional Kalman filter algorithm for the states, parameters and order of fractional system estimation, *International Journal of Applied Mathematics and Computer Science* 16(1) (2006), 129–140, full-text record.
- [2] W. Yu, Y. Zhang, R. Chen, W. Zhu, C. Wen, and Y. Chen, A fractional high-order extended Kalman filter for fractional-order nonlinear systems, *ISA Transactions* 171 (2026), 152–162, doi link.

- [3] R. Malti, T. Raïssi, M. Thomassin, and F. Khemane, Set membership parameter estimation of fractional models based on bounded frequency domain data, *CNSNS* 15 (2010), 927–938, doi link.
- [4] F. Khemane, R. Malti, T. Raïssi, and X. Moreau, Robust estimation of fractional models in the frequency domain using set membership methods, *Signal Processing* 92 (2012), 1591–1601, doi link.
- [5] N. Nazarian, M. Haeri, and M. S. Tavazoei, Identifiability of fractional order systems using input output frequency contents, *ISA Transactions* 49 (2010), 207–214, doi link.
- [6] M. S. Tavazoei and M. Haeri, A proof for non existence of periodic solutions in time invariant fractional order systems, *Automatica* 45 (2009), 1886–1890, doi link.
- [7] J.-S. Duan, Y.-J. Lan, and M. Li, A comparative study of responses of fractional oscillator to sinusoidal excitation in the Weyl and Caputo senses, *Fractal and Fractional* 6 (2022), 692, doi link.
- [8] S. Abrashov, R. Malti, M. Moze, X. Moreau, F. Aioun, and F. Guillemard, Simple and robust experiment design for system identification using fractional models, *IEEE TAC* 62 (2017), 2648–2658, doi link.
- [9] M. Karimshoushtari and C. Novara, Design of experiments for nonlinear system identification: A set membership approach, *Automatica* 119 (2020), 109036, doi link.
- [10] A. Rauh and R. Malti, Quantification of time-domain truncation errors for the reinitialization of fractional integrators, *Acta Cybernetica* 26 (2023), 105–128, doi link.
- [11] A. Rauh and M. Lahme, Observer-based error correction in finite memory approaches for pseudo state estimation of fractional dynamic models, *Annual Reviews in Control* 60 (2025), 101018, doi link.
- [12] S. Das and K. Maharatna, Fractional dynamical model for the generation of ECG like signals from filtered coupled Van-der-Pol oscillators, *Computer Methods and Programs in Biomedicine* 112(3) (2013), 490–507, doi link.
- [13] A. Takha, M. L. Talbi, and P. Ravier, Fractional calculus integration for improved ECG modeling: A McSharry model expansion, *Medical Engineering & Physics* 132 (2024), 104237, doi link.
- [14] K. Gillette et al., A framework for the generation of digital twins of cardiac electrophysiology from clinical 12-leads ECGs, *Medical Image Analysis* 71 (2021), 102080, doi link.
- [15] T. Grandits, K. Gillette, G. Plank, and S. Pezzuto, Accurate and efficient cardiac digital twin from surface ECGs: Insights into identifiability of ventricular conduction system, *Medical Image Analysis* 105 (2025), 103641, doi link.
- [16] S. M. Rump, Verification methods: Rigorous results using floating-point arithmetic, *Acta Numerica* 19 (2010), 287–449, doi link.
- [17] A. Neumaier, Rigorous sensitivity analysis for parameter-dependent systems of equations, *Journal of Mathematical Analysis and Applications* 144 (1989), 16–25, doi link.
- [18] A. Goldsztejn and L. Jaulin, Inner and outer approximations of existentially quantified equality constraints, in *CP 2006, LNCS 4204*, 198–212, doi link.
- [19] F. Johansson, Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Transactions on Computers* 66 (2017), 1281–1292, doi link.
- [20] B. Ponleitner and H. Schichl, Exclusion regions for parameter-dependent systems of equations, *Journal of Global Optimization* 81 (2021), 621–644, doi link.
- [21] H. Pollard, The completely monotonic character of the Mittag-Leffler function $E_\alpha(-x)$, *Bulletin of the American Mathematical Society* 54 (1948), 1115–1116, doi link.
- [22] T. Simon, Comparing Fréchet and positive stable laws, *Electronic Journal of Probability* 19 (2014), paper 16, 1–25, doi link.